



A. Question Marks

time limit per test: 1 second

memory limit per test: 256 megabytes

Tim is doing a test consisting of $4n$ questions; each question has 4 options: 'A', 'B', 'C', and 'D'. For each option, there are exactly n correct answers corresponding to that option — meaning there are n questions with the answer 'A', n questions with the answer 'B', n questions with the answer 'C', and n questions with the answer 'D'.

For each question, Tim wrote his answer on the answer sheet. If he could not figure out the answer, he would leave a question mark '?' for that question.

You are given his answer sheet of $4n$ characters. What is the maximum number of correct answers Tim can get?

Input

The first line contains a single integer t ($1 \leq t \leq 1000$) — the number of test cases.

The first line of each test case contains an integer n ($1 \leq n \leq 100$).

The second line of each test case contains a string s of $4n$ characters ($s_i \in \{A, B, C, D, ?\}$) — Tim's answers for the questions.

Output

For each test case, print a single integer — the maximum score that Tim can achieve.

Example

input	Copy
6	
1	
ABCD	
2	
AAAAAAA	
2	
AAAABBBB	
2	
???????	
3	
ABCABCABC	
5	
ACADC??ACAC?DCAABC?C	
output	Copy
4	
2	
4	
0	
9	
13	

Note

In the first test case, there is exactly one question with each answer 'A', 'B', 'C', and 'D'; so it's possible that Tim gets all his answers correct.

In the second test case, there are only two correct answers 'A' which makes him get exactly 2 points in any case.

Codeforces Round 963 (Div. 2)

Finished

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→ Problem tags

[greedy](#) [implementation](#) [*800](#)

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→ Contest materials

- [Announcement \(en\)](#) 
- [Tutorial #1 \(en\)](#) 
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In the third test case, Tim can get at most 2 correct answers with option 'A' and 2 correct answers with option 'B'. For example, he would get 4 points if the answers were 'AACBBDD'.

In the fourth test case, he refuses to answer any question at all, which makes him get 0 points.

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